

# Complete Terraform + Ansible Lab Guide

## End-to-End Lab with Repository Structure, Code, Commands, and Validation Steps

### GitHub Repository Structure

```
skillzrevo_devops/  
├── terraform/  
│   ├── ansible-lab/  
│   │   ├── main.tf  
│   │   ├── outputs.tf  
│   │   └── terraform.tfvars  
│   └── ansible/  
│       ├── inventory.ini  
│       ├── nginx.yml  
│       └── README.md  
├── docker/  
│   └── (future docker labs)  
└── docs/  
    └── Complete_Terraform_Ansible_Lab_With_Code.pdf
```

### Step 1 - Terraform Infrastructure

Objective: Create two EC2 instances (Control Node and Managed Node) and a Security Group.

```
resource "aws_instance" "ansible_control" {  
    ami           = "<ubuntu-ami>"  
    instance_type = "t2.micro"  
    key_name      = "ansible-lab-key"  
}  
  
resource "aws_instance" "ansible_managed" {  
    ami           = "<ubuntu-ami>"  
    instance_type = "t2.micro"  
    key_name      = "ansible-lab-key"  
}
```

### *Terraform Commands Executed*

```
terraform init  
terraform fmt  
terraform validate  
terraform plan  
terraform apply
```

### *Outputs Used*

```
terraform output  
  
control_public_ip = 44.xxx.xxx.xxx  
managed_public_ip = 13.xxx.xxx.xxx  
managed_private_ip = 172.31.xx.xx
```

## Step 2 - SSH Access

```
ssh -i /c/Users/aws_keys/ansible-lab-key.pem ubuntu@44.xxx.xxx.xxx
```

## Install Ansible on Control Node

```
sudo apt update
sudo apt install ansible -y
ansible --version
```

## Step 3 - Create Inventory

```
mkdir ~/ansible-lab
cd ~/ansible-lab

nano inventory.ini
[web]
172.31.xx.xx

[web:vars]
ansible_user=ubuntu
```

## Step 4 - Copy SSH Key

```
scp -i ansible-lab-key.pem ansible-lab-key.pem ubuntu@44.xxx.xxx.xxx:/home/ubuntu/
chmod 400 ansible-lab-key.pem
```

## Step 5 - Verify Connectivity

```
ansible -i inventory.ini web -m ping
SUCCESS => {
  "ping": "pong"
}
```

## Step 6 - Ad-Hoc Commands

```
ansible -i inventory.ini web -m apt -a "name=nginx state=present" -b
ansible -i inventory.ini web -m service -a "name=nginx state=started enabled=yes" -b
ansible -i inventory.ini web -m copy -a 'content="Hello from Ansible Lab" dest=/var/www/html/index.h'
```

## Validation

```
curl http://172.31.xx.xx

Output:
Hello from Ansible Lab
```

## Step 7 - Nginx Playbook

```
---
- name: Install and Start Nginx
  hosts: web
  become: yes

  tasks:

    - name: Install nginx
      apt:
        name: nginx
        state: present
        update_cache: yes

    - name: Start nginx service
      service:
        name: nginx
        state: started
        enabled: yes

    - name: Deploy custom homepage
      copy:
        content: "Hello from Ansible Playbook"
        dest: /var/www/html/index.html
```

## Execute Playbook

```
ansible-playbook -i inventory.ini nginx.yml
```

## Verify Service

```
ansible -i inventory.ini web -m shell -a "systemctl status nginx --no-pager" -b
```

## Verify Content

```
curl http://172.31.xx.xx
```

Output:  
Hello from Ansible Playbook

## Key Interview Concepts Learned

1. Terraform = Infrastructure Provisioning
2. Ansible = Configuration Management
3. Inventory = Target Hosts Definition
4. Playbook = Reusable Automation
5. Ad-Hoc Commands = One-Time Tasks
6. Idempotency = Same Result Every Time
7. SSH = Ansible Communication Method
8. Control Node vs Managed Node
9. Service Validation
10. Infrastructure as Code + Configuration as Code

## Next Session: Docker Fundamentals

Containers vs VMs  
Docker Architecture  
Images vs Containers  
Dockerfile  
Build and Run Containers  
Container Security Basics